Action, Trajectories and Gaps: A Research Programme

navstokgap research working notes

Abstract

We organize the accepted action-scale results around the physical premises that supply positivity and the counterexamples a selection principle must exclude. Quantum reconstruction and interacting spectral control are the main research tracks. The programme identifies their next proof obligations and keeps dimensional action selection, calibrated observable access and uniform limits explicit. The maintained source is `research/PROGRAMME.md`; `research/STATE.md` supplies the single current priority list.

1 Research programme: quantum necessity and robust gap mechanisms

Version 53, 2026-09-13. The programme consolidates the accepted results into Classical action scales: obstructions, conditional bounds and quantum premises. The user's 2026-09-12 redirection makes quantum-premise selection and interacting gap control the main tracks. STATE owns the current ordered tasks; STRATEGY governs bounded choices and stopping decisions.

1.1 Aim and present conclusion

Derive the need for quantum structure with a positive universal action parameter from explicit physical premises, and develop a precise mechanism for a gap that survives the relevant limits. A construction admitting quantum inputs and a construction selecting those inputs answer different questions. Newtonian cut refinement remains the consistency test for proposed action mechanisms; the Millennium comparison supplies the field-theory target.

The accepted examples provide a substantial set of obstructions to simple classical selection arguments and several conditional positive bounds. The physical premise excluding their quiet, contracting or precisely observed countermodels remains open. Spectral work gives finite-system criteria, independent-product control and C124's interacting Ising gap uniform in finite volume at bounded coupling. Physical-operator identification and further uniform limits remain open.

This version integrates G03's interacting finite-volume gap C124 and its bounded B68 source audit. No novelty is asserted. The ledger retains proof and literature status, including the limits of each source audit.

1.2 Quantum track: identify the excluded classical alternative

Q01's first milestone compares Hardy and Chiribella–D'Ariano–Perinotti through the exact premise that rules out a classical operational model. The bounded premise map and B67 audit are complete: Hardy requires continuous reversible pure-state connectivity; CDP requires pure extensions with essential uniqueness. Standard finite classical models fail these premises. Hardy's moving-ball example points to the extra question of an exact finite operational description closed during transformations. Neither full reconstruction proof has been audited here. For each selected passage specify the state space, preparations, transformations, composition and measurement assumptions. Test the classical alternative in that same class and mark the step where action units would still need to enter.

K01 is the supporting fixed-time compatibility test. It asks whether local measurement descriptions extend to one system-plus-apparatus state and which assumption about context independence changes the answer. A primary Kochen–Specker comparison is a source task with explicit hypotheses. Merely hiding apparatus variables is not enough to exclude their classical model. I007 records the construction.

The B01 companions provide the existing Hardy/CDP source routes and limited reading coverage. Full reconstruction-proof acceptance remains future work. The historical source-to-model connections are preserved under their own tasks and can supply a specific premise or example.

1.2.1 Action selection obligations

A candidate must discharge five gates in the action target: classical definition and units; exclusion of zero; convergence in a named topology; universality across the admitted masses and preparations; and identification with quantum phase or commutation structure. Define $\mathsf{h}_\varepsilon(t)$ when using the proposed action-field route, keeping physical time t and resolution epsilon distinct. Observation duration Delta, volume and c^{-1} have their own limits.

The checkerboard calculation takes a supplied action parameter, amplitudes and measurement rule to a coherent Dirac limit. It is the Branch A comparison for a Branch B necessity argument. A state-space reconstruction still needs a mechanical dynamical identification of its dimensional scale; numerical calibration is a further empirical task.

1.3 Gap track: interactions and uniform control

G01/G02 establish the finite reversible-generator estimate

$$\gamma \geq \frac{\alpha}{S}, \qquad S = \sum_a \langle f_a, (-Q)^{-1} f_a \rangle,$$

where alpha is the lower observable-frame bound on centered states. One positive measured plateau can miss an arbitrarily slow mode; exact state identification can also become poorly conditioned. Independent product dynamics control mixed modes and give the minimum constituent gap when each constituent clock is retained (C041–C046).

G03 now supplies an explicit interacting estimate: the periodic zero-field Ising heat-bath chain has full gap $a[1 - \tanh(2b)]$ for every $N \geq 3$. Here a is the per-site refresh rate and $b \geq 0$ the dimensionless nearest-neighbour coupling. Hamming contraction controls all modes and magnetization attains the bound. Bounded coupling and a clock floor give a uniform finite-volume gap; finite range alone permits joint coupling/size closure, and a fixed total clock adds a $1/N$ slowdown. The C124 proof replaces tensorization without an extensive full-frame susceptibility sum. B68 records a bounded model/method audit, including unavailable Glauber formula images. Further interaction variants are parked until a named operator or limit dependency requires them. See also the susceptibility proof.

A Yang–Mills transfer additionally requires its physical quantum Hamiltonian, gauge-invariant observables, continuum and infinite-volume construction and quantum axioms. An auxiliary sampling generator's relaxation rate changes under clock rescaling while its invariant law stays fixed. The comparison note and official-target digest preserve these distinctions. A finite-box decay estimate or a toy Hessian gap supplies only its own operator statement.

1.4 Consolidated evidence and retained supporting models

Mechanism tested	Accepted consequence	Premise still requiring selection
Continuous variation and scale contraction	C002/C056–C058 give zero action infima in specified classes	Restriction excluding the allowed variations or low excitation
Positive orbit cost	C060–C061 give a sharp force/speed-floor bound	Physical origin and universality of the excitation floor
Fluctuation and independent composition	C035–C036 give a common coefficient using a positive reference	Preparation equivalence and origin of reference fluctuations
Prepared apparatus and records	C068–C123 give recovery and ambiguity under explicit data restrictions	Irreducible, universal information/preparation constraints
Coherent paths	C039–C040 give continuum and measured-cut comparisons	Necessity of amplitudes, measurement rule and action parameter
Observable response and interaction	C041–C046 give gap criteria; C124 gives an interacting uniform finite-volume gap	Origin of coupling/clock control, physical operator identification and further limits

The consolidated paper supplies short arguments and links to the full proofs. The receiver/calibration compilation retains technical details, including R33's open curvature test. R33 is parked: its next coefficient currently changes no selected quantum or gap premise. It can return for a named dependency, proof correction or an explicit user request.

A19's peak-excitation estimate, A14's renewal-preparation model, R02's repaired finite-propagation drafts and M03's spectral laboratory remain supporting work. They become selected work when their possible outcomes change a main premise decision. New solvable subcases do not automatically displace the main tracks.

1.5 Cut refinement and historical context

The cut-point target collects the existing consistency tests. Constant-force chord error closes under refinement; fixed Gaussian bridge laws preserve their supplied fluctuation parameter; velocity and receiver memory distinguish retained-state cuts from interventions. Joint path/action limits require their stated topology and preparation changes. These results constrain proposed mechanisms without selecting quantum weights.

Newton's geometric arguments, the cone/arrow proposals, the Classical Scholia and the H10 static-composition readings remain source-backed research inputs. The bibliography bridge index routes a named task to a construction, premise or exact passage. H11 strengthens source witnesses when selected; it does not resume merely because an old source card calls it next. The supplied bibliography note remains intact, with its separate mathematical selection.

1.6 Acceptance, cost and consolidation

Use written proofs and bounded source review. No numerical or symbolic verification scripts may be created or run in Python or another language. Historical scripts and outputs remain records. Document/source/integrity Python is allowed. Delegation is selective and sequential under the protocol, with one Sol/Luna worker and explicit supported effort, never ultra.

Before each step, record its target obligation, the decision changed by its possible outcomes, its deliverable/stop and why existing results are insufficient. After two consecutive steps in a proof family, compare its continuation with the other main track. The coordinator makes that choice without routine user approval. Correct substantive errors when found and preserve explicit user scope. A completed local calculation need not spawn another calculation.

A new result receives written proof review and a bounded librarian audit. Consolidation reuses those audits with an evidence map; it does not imply a fresh prior-art search or automatic novelty. Keep STATE short and reconcile its queue with TASKS and the latest handoff. Dated milestones live in the state history and individual handoffs.

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